SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I K Phani Sridhar Yogi (192311091) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

K.P.Sridhar

Signature of the student:

1. RSA Key Generation and Security Analysis

Given:

p = 17, q = 19

Compute n and ϕ(n)

Choose e = 5 and calculate the private key d

Encrypt the message M = 7

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

Solution :

1. RSA Key Generation and Security Analysis

Given:

- $p = 17$
- $q = 19$
- $e = 5$
- Message $M = 7$

Step 1: Compute $\boldsymbol{n}$and $\boldsymbol{\phi(n)}$

$$n = p \times q = 17 \times 19 = 323$$

Euler's Totient:

$$\phi(n) = (17 - 1)(19 - 1) = 16 \times 18 = 288$$

Answer

- $n = 323$
- $\phi(n) = 288$

Step 2: Find Private Key $\boldsymbol{d}$

Need

$$d \times e \equiv 1 \pmod{288}$$

So,

$$5d \equiv 1 \pmod{288}$$

Trying multiples:

$$5 \times 173 = 865$$
$$865 \div 288 = 3 \text{ remainder } 1$$

Hence,

$$d = 173$$

Step 3: Encrypt Message

Formula:

$$C = M^e \ n$$
$$C = 7^5 \ 323$$

Calculate:

$$7^2 = 49$$
$$7^4 = 49^2 = 2401$$
$$2401 \ 323 = 140$$
$$7^5 = 140 \times 7 = 980$$
$$980 \ 323 = 11$$

Ciphertext

$$\boxed{C = 11}$$

Step 4: Decrypt

Formula

$$M = C^d \ n$$
$$M = 11^{173} \ 323$$

Using modular exponentiation,

$$11^{173} \ 323 = 7$$

Recovered message

$$\boxed{M = 7}$$

The decrypted message matches the original.

Security Analysis

Using very small prime numbers makes RSA insecure because:

- Small keys are easy to factor.
- An attacker can quickly determine $p$and q.
- Once $p$and $q$are known, the private key can be calculated.
- Modern RSA uses **2048-bit or larger keys** for strong security.

2. ElGamal Encryption Analysis

Given:

p = 23, g = 5, private key x = 7, random k = 3, message M = 10

Compute the public key

Perform encryption to generate ciphertext pair (C1, C2)

Decrypt the ciphertext to retrieve the original message

Analyze how randomness (k) improves security in ElGamal

Solution :

Step 1: Compute Public Key

Formula

$$y = g^x \ p$$
$$y = 5^7 \ 23$$

Calculate:

$$5^2 = 25 \equiv 2$$
$$5^4 = 2^2 = 4$$
$$5^7 = 5^4 \times 5^2 \times 5$$
$$= 4 \times 2 \times 5 = 40$$
$$40 \ 23 = 17$$

Public key

$$\boxed{(23, 5, 17)}$$

Step 2: Encryption

Compute C_1

$$C_1 = g^k \ p$$
$$= 5^3 \ 23$$
$$125 \ 23 = 10$$

Compute C_2

$$C_2 = M \times y^k \ p$$

First

$$17^2 = 289 \ 23 = 13$$
$$17^3 = 13 \times 17 = 221$$
$$221 \ 23 = 14$$

Now

$$10 \times 14 = 140$$
$$140 \ 23 = 2$$

Ciphertext

$$\boxed{(10, 2)}$$

Step 3: Decryption

Receiver computes

$$s = C_1^x \; p$$
$$10^7 \; 23 = 14$$

Inverse of 14 modulo 23 is

$$14^{-1} = 5$$

Recovered message

$$M = C_2 \times 14^{-1} \; 23$$
$$= 2 \times 5 = 10$$

Recovered message

$$\boxed{M = 10}$$

Security Analysis

Random value k:

- Produces different ciphertexts for the same message.
- Prevents replay and pattern analysis.
- Makes ElGamal probabilistic.
- Greatly increases security against attackers.

3. RSA Digital Signature Verification

Given:

p = 11, q = 17, e = 7, message M = 8

Compute n and ϕ(n)

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

Solution :

Given

- $p = 11$
- $q = 17$
- $e = 7$

- $M = 8$

Step 1: Compute n

$$n = 11 \times 17 = 187$$

Step 2: Compute Totient

$$\phi(n) = 10 \times 16 = 160$$

Step 3: Find d

Need

$$7d \equiv 1 \pmod{160}$$
$$7 \times 23 = 161$$
$$161\ 160 = 1$$

Therefore

$$\boxed{d = 23}$$

Step 4: Generate Signature

Formula

$$S = M^d\ n$$
$$S = 8^{23}\ 187$$

Using modular exponentiation,

$$\boxed{S = 173}$$

Step 5: Verify Signature

Receiver computes

$$V = S^e\ n$$
$$173^7\ 187 = 8$$

Recovered value equals the original message.

Hence,

$$\boxed{\text{Signature Verified}}$$

Non-Repudiation Analysis

Digital signatures ensure:

- Authentication of the sender.
- Integrity of the message.
- Non-repudiation because only the private key owner could have generated the signature.
- Any modification to the message causes verification to fail.

4. Diffie-Hellman with Attack Resistance Evaluation

Given:

p = 29, g = 2

User A private key = 5

User B private key = 12

Compute public keys for both users

Compute the shared secret key

Analyze what happens if an attacker intercepts and replaces public keys

Suggest a cryptographic solution to ensure authenticity

Solution :

Given

- $p = 29$
- $g = 2$
- User A private key = 5
- User B private key = 12

Step 1: Public Keys

User A

$$A = 2^5 \ 29$$

$$32 \ 29 = 3$$

User B

$$B = 2^{12} \ 29$$
$$2^{12} = 4096$$
$$4096 \ 29 = 7$$

Public keys

- A = 3
- B = 7

Step 2: Shared Secret

User A

$$K = 7^5 \ 29$$
$$= 16$$

User B

$$K = 3^{12} \ 29$$
$$= 16$$

Shared key

$$\boxed{16}$$

Attack Analysis

If an attacker intercepts the exchanged public keys and replaces them:

- User A unknowingly establishes a secret with the attacker.
- User B also establishes a different secret with the attacker.
- The attacker decrypts, reads, modifies, and re-encrypts all communication.
- This is known as a Man-in-the-Middle (MITM) attack.

Cryptographic Solution

To prevent MITM attacks:

- Use digital signatures to authenticate public keys.
- Use digital certificates issued by a trusted Certificate Authority (CA).
- Protocols such as TLS/SSL combine Diffie–Hellman with authentication to ensure both secrecy and authenticity.

5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption

RSA encryption time = 5 ms per key exchange

AES encryption time = 0.5 ms per data block

Total data blocks = 500

Calculate total time taken for encryption

Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient

Justify its use in modern secure communication systems

Solution :

Given

- RSA key exchange time = **5 ms**
- AES encryption = **0.5 ms per block**
- Total blocks = **500**

Step 1: Hybrid Encryption Time

AES encryption time

$$500 \times 0.5 = 250 \text{ ms}$$

Add RSA key exchange

$$250 + 5 = 255 \text{ ms}$$

Total Hybrid Time

$$\boxed{255 \text{ ms}}$$

Step 2: RSA Only

RSA encrypts every data block.

$$500 \times 5 = 2500 \text{ ms}$$

Total RSA Time

$$\boxed{2500 \text{ ms}}$$

Step 3: Comparison

Method	Total Time
Hybrid (RSA + AES)	**255 ms**
RSA Only	**2500 ms**

Hybrid encryption is approximately:

$$\frac{2500}{255} \approx 9.8$$

about **10 times faster**.

Analysis

Hybrid encryption is more efficient because:

- RSA is computationally expensive and is best suited for encrypting small amounts of data, such as symmetric keys.
- AES is a fast symmetric-key algorithm designed for encrypting large volumes of data efficiently.
- Using RSA only once to exchange the AES key, followed by AES for bulk data encryption, provides both security and high performance.

Justification for Modern Secure Communication

Hybrid encryption is widely used in secure communication protocols because it combines the strengths of both cryptographic approaches:

- **RSA** securely exchanges encryption keys.
- **AES** provides fast encryption and decryption of large amounts of data.
- The approach offers confidentiality, efficiency, scalability, and secure key management.
- Modern protocols such as **TLS/SSL**, **HTTPS**, **VPNs**, and secure messaging applications use hybrid encryption to achieve strong security with minimal performance overhead.